



***BASE24-pos Visa II
Device Support Manual***

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Contents

Preface.....	ix
Conventions Used in this Manual.....	xiii

Section 1

Introduction.....	1-1
BASE24-pos VISA II Device Handler Module Overview	1-2
Supported Terminals	1-3
Supported Protocols	1-4
Supported Features	1-5
Expanded Transaction Set	1-6
Multiple Terminal Support.....	1-6
Multiple Language Support.....	1-6
VISA External Message Formats	1-6
Draft Capture Option	1-7
BASE24-pos Settlement and Cutover	1-7
Terminal Balancing	1-8
BASE24 Transaction Security.....	1-8
Sequence Number Checking	1-8
Log Message Generation.....	1-8
PS2000 Support.....	1-9
Track 1 and Track 2 Support.....	1-9
Address Verification.....	1-9
Extended Address Verification.....	1-9
Port Usage Tracking	1-10
Gift Card and Prepaid Card Transactions.....	1-10
Visa Card-Level Results.....	1-10
Contactless Transactions	1-11
VVDH Transaction Support.....	1-12
Financial Transactions.....	1-12
Administrative Transactions.....	1-14
Integration with BASE24-pos	1-15
Transaction Flow	1-15
Tying the VVDH Module to BASE24-pos	1-16

Section 2

Terminal Configuration	2-1
VVDH Configuration Files	2-2
POS Retailer Definition File (PRDF).....	2-2
POS Terminal Data File (PTDF).....	2-2
POS Transaction Log File (PTLF)	2-2

Logical Network Configuration File	2-3
Assigns	2-3
Params	2-3
Section 3	
VISA II Device Handler (VVDH) Module	3-1
VVDH Module Overview	3-2
Enforcing Transaction Sequence	3-2
Updating Accumulator Fields and Status Flags	3-2
Cutting Over Terminals	3-2
Decrypting PINs	3-3
Controlling Response Timers	3-3
File Usage	3-4
VVDH Module Initialization and Warmboot	3-5
Initializing the VVDH Module	3-5
Warmbooting the VVDH Module	3-7
Network Text Commands	3-8
Log Message Generation	3-9
Section 4	
VISA II External Message	4-1
Standard Message Headers	4-2
Record Format Element	4-2
Application Type Element	4-3
Optional Data	4-4
Request message requirements	4-6
Response message requirements	4-7
Transaction codes	4-8
Section 5	
VVDH Configuration Considerations	5-1
PIN Encryption	5-2
Sequence Number Checking	5-3
Draft Capture	5-4
Authorization with Electronic Follow-up	5-4
Setting the Draft Capture Flag in the PSTM	5-5
Transaction Accumulation Totals	5-6
Section 6	
Transaction Message Flows	6-1
Approved Normal Purchase Received by Device	6-2
Approved Normal Purchase; Communications between Device and BASE24 Down	6-3
Declined Normal Purchase Received by Device	6-5
Declined Normal Purchase; Communications between Device and BASE24 Down	6-6
Batch Close Settlement Transaction - Good Batch	6-8
Batch Close Settlement Transaction - Rejected Batch	6-10

Appendix A	
American Express Standard Industry Formats	A-1
Index	Index-1

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Preface

The *BASE24-pos VISA II Device Support Manual* is designed to provide readers with a thorough understanding of the BASE24-pos VISA II Device Handler (VVDH) module. It presents readers with an extensive overview of the VVDH module and its options, along with providing information on how to integrate the VVDH module for use with BASE24-pos. In addition, this manual describes, in detail, the processing logic and configuration issues associated with the VVDH module.

Audience

There are three targeted audiences of this manual. One audience reads this manual to get a general overview of the versatility of the VVDH module and terminals that meet its specifications. Another audience consists of terminal vendors who want to configure terminals that are compatible with the VVDH module. The third audience of this manual is customers who are purchasing the VVDH module and contracting with terminal vendors.

Prerequisites

Although knowledge of BASE24-pos is helpful for the customer who is reading this manual, the manual is designed to stand alone. Terminal vendors and personnel who seek an overview of the VVDH module should be able to find as much information about BASE24-pos as they require to perform their functions. However, if more information about BASE24-pos is desired, readers should refer to the *Introduction to BASE24-pos* and the *BASE24-pos Transaction Processing Manual*.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- ◆ The *BASE24 Base Files Maintenance Manual* discusses the Base files used with the VVDH module.

- ◆ The ***BASE24 CRT Access Manual*** contains general information on logging on to the BASE24 system, using the default function keys, and creating screen access security.
- ◆ The ***BASE24 Logical Network Configuration File (LCONF) Manual*** discusses in detail the assigns and params used in configuring the VVDH module.
- ◆ The ***BASE24 Log Message Reference Manual*** discusses in detail how to access log messages generated by the VVDH module.
- ◆ The ***BASE24 Tokens Manual*** discusses the American Express token and Check Guarantee/Verification token.
- ◆ The ***BASE24 Text Command Reference Manual*** contains detailed descriptions and syntax of all text commands.
- ◆ The ***BASE24-pos Router/Authorization Manual*** discusses the steps taken by the Router and Authorization modules when initializing.
- ◆ The ***BASE24-pos Settlement and Reporting Manual*** discusses in detail BASE24-pos settlement, cutover, balancing, and reporting.
- ◆ The ***BASE24-pos Transaction Processing Manual*** discusses in detail transaction processing used in configuring the VVDH module.
- ◆ The ***BASE24-pos Transaction Security Manual*** discusses in detail PINs, message authentication codes (MACs), and dynamic key management used to set up transaction security for the VVDH module.
- ◆ The ***BASE24-pos Files Maintenance Manual*** discusses the BASE24-pos files used with the VVDH module.

This manual also contains references to the ***NET24-XPNET Network Control Command Reference Manual*** which discusses in detail the process, station, line and device declarations used in configuring the VVDH module.

Software Release

This manual documents standard processing for BASE24-pos Release 6.0.

Customer-specific code modifications (that is, CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

Section 1, "Introduction," provides an overview of the VVDH module and its functionality. This section includes information explaining how the VVDH module can

be integrated with BASE24-pos. This section also discusses the role of the VVDH module during transaction processing.

Section 2, "Terminal Configuration," explains the options available to customers configuring their POS terminals. This section explains the files that must be set up in order to tailor VVDH module configuration options to meet the needs of a particular customer. This section includes descriptions of the Logical Network Configuration File (LCONF) assigns and params used by the VVDH module.

Section 3, "VISA II Device Handler (VVDH) Module," contains information about the VVDH module. It includes an overview of the module, a description of each file used by the VVDH module during processing, a description of the steps the VVDH module performs during initialization and warmbooting, and a list of the network control commands recognized by the VVDH module.

Section 4, "VISA II External Message," contains information about the message formats used by the VVDH module to communicate with terminals.

Section 5, "VVDH Configuration Considerations," presents information on topics the customer must consider when deciding how to configure the VVDH module. It includes PIN encryption, sequence number checking, draft capture options, and transaction accumulation totals information.

Section 6, "Transaction Message Flows," contains transaction flows between the terminal and the VVDH module.

Appendix A, "American Express Standard Industry Formats," defines the VVDH module's use of the American Express standard industry formats. Generally, American Express has defined categories under which transactions are placed. For each of these categories, American Express requires different data to be sent from the device. Therefore, American Express has set forth standard industry formats for each category to ensure the data they require is captured by the device and sent to BASE24.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, PS-DP240-01 for the ***BASE24-pos VISA II Device Support Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 10/2009 for October, 2009) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions. The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record

entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

Conventions Used in this Manual

This section discusses module terminology, explains how unlabeled fields on screens are documented in this manual, and also explains a change in Nucleus process terminology.

Module Terminology

The VISA II Device Handler module is comprised of three components:

- Main processing module
- Data Capture Upload module (optional)
- Load Library module

The three together comprise what is referred to in this document as the VVDH module.

Unlabeled Fields on Screens

Angle brackets (<>) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

Unlabeled fields are not included in the index by field name; they appear by subject in the main index.

Nucleus Process Terminology Change

There is one process that is the hub of an XPNET system. It manages messages queues; controls all lines, stations, and processes in the system; and serves as an interprocess interface. Throughout this manual, this process is referred to as the XPNET process.

In other manuals, you may see references to the Nucleus process as performing these same functions. These two terms (Nucleus process and XPNET process) should be viewed as synonymous. The Nucleus process and the XPNET process perform the same functions

and processing. As manuals are reissued, references to the Nucleus process will be replaced with references to the XPNET process.

Section 1

Introduction

This section contains introductory-level information describing the BASE24-pos VISA II Device Handler (VVDH) module and the functions it can perform. The main purpose of this section is to provide readers with the background information they require as a basis to build a solid foundation for understanding the VVDH module. This section introduces the main topics important when using the VVDH module and provides general overviews for each specific topic identified.

This section is intended for readers who require only an overview of the VVDH module and its functionality. It provides information on topics important in gaining a basic understanding of the VVDH module, such as supported features and terminals. However, this section is also helpful to readers who require a detailed explanation of the VVDH module and the functions it can perform. This section lays the groundwork to prepare these readers for information covered in the rest of the manual by providing them with previews of topics discussed in greater detail in subsequent sections.

This section also includes the following topics concerning the VVDH module:

- Supported terminal models
- Supported protocols
- Supported features
- Supported transactions
- Integration with BASE24-pos
- VVDH module version information

BASE24-pos VISA II Device Handler Module Overview

The VISA II Device Handler (VVDH) module operates in conjunction with the BASE24-pos product. BASE24-pos is a point-of-sale transaction processing system. The VVDH module provides flexible transaction support, configuration options, and allows retailers to select the functionality to best meet their particular business requirements.

The VVDH module can handle processing for POS transactions sent from a terminal. Among others, these include Purchase, Purchase Adjustment, Cash Advance, and Merchandise Returns. A complete list of transaction types supported by the VVDH module is included later in this section. The VVDH module also offers flexible security options to ensure that transactions are processed only once and various PIN encryption techniques.

Supported Terminals

The VISA II Device Handler (VVDH) module provides support for the following terminals:

- VISA II Dial terminal

The VVDH module can communicate with any POS terminal having the capability of sending messages to and receiving messages from the VVDH module using the VISA Second Generation record formats. Terminals must be able to communicate with the VVDH module and allow the VVDH module to communicate with them. Terminals can communicate with the VVDH module over dial lines, leased lines, or a communications protocol network.

Supported Protocols

The VVDH module can be configured to use the following communication protocols:

- VISA
- VISA 1
- VISA 2
- X.25

The protocol selected by the customer depends on the line communications environment established between the VVDH module and its terminals. Any protocol that requires protocol-specific processing on the part of the VVDH module is not supported.

Authorized ACI personnel assist customers at installation or when a system is configured in determining the type of protocol best suited for the customer's particular requirements. For detailed information on protocols and their usage, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

Supported Features

An enhanced set of features is available for use with the VVDH module. Customers can select and configure the features necessary to meet their processing requirements. This subsection provides a list of these features, followed by a summary of each feature. The following features are supported by the VVDH module:

- Expanded transaction set
- Multiple terminal support
- Multiple language support
- VISA external message formats
- Draft capture options
- American Express Data Collection (limited usage)
- BASE24-pos settlement and cutover support
- Terminal balancing
- BASE24 transaction security
- Sequence number checking
- Log message generation
- PS2000 support
- Track 1 and Track 2 Support
- Address Verification
- Extended Address Verification
- Port Usage Tracking
- Gift Card and Prepaid Card Transactions
- Visa Card-Level Results
- Contactless Transactions

Expanded Transaction Set

The VVDH module can handle processing for a number of POS transactions sent from a terminal. These include Purchase, Cash Advance, Check Guarantee, Check Verification, and Merchandise Returns. A complete list of the transaction set supported by the VVDH module is included later.

Multiple Terminal Support

Customers can use any POS terminal that conforms to the VVDH module requirements set forth in this manual.

Multiple Language Support

Customers have the option of sending responses to a terminal in one of three different languages. Terminal responses are displayed in the language set up in the LANGUAGE-IND field in the PTDF. The LANGUAGE-IND field identifies the language table used by the VVDH module when sending responses to the terminal. Responses can be displayed on the terminal in one of three languages. The language tables exist as arrays created in edit files. The edit files are then sourced into the VVDH module. The particular language represented in each language table is determined by the customer. For example, language table 1 can be English, language table 2 can be French, and language table 3 can be Spanish. For more information on the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***.

VISA External Message Formats

The VVDH external message is a variable-content, variable-length message, based on the message formats developed by Visa International. VISA Second Generation message formats supported are:

- Format 5 - Initial Second Generation Terminal format
- Format F - Second Generation Terminal format superseding Format 5 (non-REPS CVV)
- Format J - Credit Card and Check Guarantee (PS/2000 REPS)
- Format G - Direct Debit authorization (PS/2000 REPS)

- Format K - Credit Card and Check Guarantee (PS/2000 CPS)
- Format A - Direct Debit authorization
- Format C - Direct Debit confirmation
- Format D - Credit Card addendum
- Format T – Debit Cards or Electronic Benefits Transfer (EBT)
- Format V – Debit Cards or Electronic Benefits Transfer(EBT) Confirmation

Draft Capture Option

The draft capture option common for second generation terminals consists of authorizing the transaction online and then, sending a separate transaction at the end of a batch to verify that the transaction was completed correctly. In effect, this consists of two transactions for every one purchase.

BASE24-pos Settlement and Cutover

BASE24-pos provides an efficient method of settlement and cutover. *Settlement*, in terms of BASE24-pos, is actually a reconciliation effort whereby the current day's transaction activity is closed out and reported, and the system is shown to be in balance. The term *cutover* simply refers to the point in time where transactions stop being logged to the current day's business and start being logged to the next day's business. It refers to the closing of the current posting day and the opening of the next.

BASE24-pos settlement differs from the settlement carried out by participating BASE24-pos financial entities (for example, retailers, retailer sponsors, cardholders, card issuers, interchanges) in that BASE24-pos does not move funds to settle accounts. It does, however, provide the support that allows these financial entities to initiate the movement of funds. An extensive set of reports provides financial entities with the information they require to settle their accounts. BASE24-pos also provides the POS Transaction Log File (PTLF), which keeps a record of each transaction processed by BASE24-pos during its defined reporting day. In addition, BASE24-pos provides an automated method to cut over terminals, retailers, institutions, interchanges, and the BASE24-pos network to a new posting day. For more information on BASE24-pos settlement, cutover, and reporting, refer to the ***BASE24-pos Settlement and Reporting Manual***.

Terminal Balancing

The VVDH module allows merchants to balance their terminals with BASE24-pos. The terminal retains totals independent of BASE24-pos and, therefore terminal totals can be compared to the BASE24-pos totals in the PTDF to determine if the totals balance.

Merchants can contact the service provider to receive terminal balancing information. The BASE24-pos report that contains terminal balancing information is the Retailer Reconciliation Report (B24POS35). This report is used to balance transaction activity at the terminal batch level and can be used by customers to determine whether their terminals balance with BASE24-pos.

BASE24 Transaction Security

The VVDH module offers various ways of ensuring that transactions are transmitted and secured correctly. These include controls that ensure every transaction is being processed only once and options that allow customers to select software PIN encryption.

For more information on BASE24 transaction security, refer to the ***BASE24 Transaction Security Manual***.

Sequence Number Checking

The VVDH module supports a method of ensuring that every message being sent from the terminal is being received only once. This method is known as sequence number checking. Sequence number checking is intended to ensure that BASE24-pos has received and processed every transaction only once. Transaction Sequence Number, in the incoming VISA message is used to detect duplicate transactions.

Log Message Generation

The VVDH module generates log messages that keep operators aware of activity occurring in the system. Log messages can be used to assist operators with their daily tasks or to notify operators of problems occurring in the system. Log messages provide operators with insight as to how the system is running.

In addition, ACI has developed online log message documentation. This online log message documentation provides customers with greater flexibility by allowing them to print out log message documentation as needed at their location or to electronically manipulate the data to suit their needs. The online log message documentation provides the message number, severity, and text for each log message generated by the VVDH

module. In addition, it also provides the cause of each message and the recommended action to take.

For complete online log message documentation, refer to the ***BASE24 Log Message Documentation***.

PS2000 Support

The VVDH module is compatible with Visa Payment Service 2000 (PS2000). PS2000 is a Visa program that ensures that the portion of the transaction that is authorized is the same as the portion that is cleared and settled. All phases of the transaction are linked, allowing the institution to reduce back office expenses.

Track 1 and Track 2 Support

The VVDH module formats Track 1 / Track 2 information from the native message into the BASE24-pos Standard Internal Message (PSTM). The VVDH module fills the Track 2 Data field of the PSTM regardless of the type of track data available in the native message. When Track 1 data is available, the VVDH module adds the Track 1 token to the PSTM.

Address Verification

The Release 6.0 VVDH module is compatible with the BASE24-pos add-on Address Verification. Address verification assists merchants in controlling losses that can result from credit card fraud during transactions where the cardholder cannot be readily identified. The VVDH module places address verification information in the PSTM. See section 4 for information about the data fields used to format address verification data in the PSTM. For more information on the BASE24-pos add-on Address Verification module, refer to the ***BASE24-pos Address Verification Manual***.

Extended Address Verification

The VVDH module supports Extended Address Verification Data. A portion of the data (Cardholder Billing Street Address and Cardholder Billing Postal Code) is placed in the Address and Zip Code fields of the request PSTM. The remainder of the data is placed in the AMEX Private Use Data Token (CX).

Port Usage Tracking

The VVDH module can track station or port usage. Port usage tracking gives an institution the ability to analyze, control, and balance network service usage. The VVDH module retrieves the station ID from the network message header and places it in the Station ID token. The token is appended to the PSTM and can be configured to be logged to the POS Transaction Log File (PTLF). Refer to the ***BASE24 Tokens Manual*** for more information on the Station ID token.

Gift Card and Prepaid Card Transactions

The VVDH module supports Gift Card and Prepaid Card Transactions. These transactions integrate with the Stored Value functionality that currently exists in BASE24-pos.

Visa Card-Level Results

The VVDH module is able to receive Visa's Card-Level Results and return them to the terminal. The following table shows the Card-Level Result values.

Note: ^ = blank character

Value	Description	Value	Description
A^	Visa Traditional	U^	Visa TravelMoney
B^	Visa Traditional Rewards	W^ - Z^	Reserved
C^	Visa Signature	0^ - 9^	Reserved
D^	Visa Infinite	G1	Visa Signature Business
E^ - F^	Reserved	G2	Visa Business Check Card
G^	Visa Business Card	J1	Visa General Prepaid
H^	Visa Check Card	J2	Visa Prepaid Gift Card
I^	Visa Commerce	J3	Visa Prepaid Healthcare
J^	Reserved	J4	Visa Prepaid Commercial
K^	Visa Corporate Card	K1	Visa GSA Corporate T&E
L^	Reserved	Q1	Private Label Prepaid
M^	MasterCard/Eurocard and Diners	S1	Visa Purchasing with Fleet
N^ - P^	Reserved	S2	Visa GSA Purchasing
Q^	Private Label	S3	Visa GSA Purchasing with Fleet
R^	Proprietary Card	V1	Reserved
S^	Visa Purchasing Card	AX	American Express
T^	Reserved/Interlink	DI	Discover

Contactless Transactions

The VVDH module supports contactless transactions from terminals. Contactless transactions are transactions initiated without physical contact between the card and terminal. Contactless transactions are supported in record formats "D" and "T".

VVDH Transaction Support

All of the transactions supported by the VVDH module must originate with a request from the terminal. A response is sent from the VVDH module for each request received unless the VVDH module is unable to read the POS Terminal Data File (PTDF) record that corresponds to the terminal sending the request. In this case, no response is sent and the terminal times out. Message requests and responses are explained in more detail later.

The VVDH module supports both financial and administrative transactions. Financial transactions are monetary transactions performed on behalf of a cardholder, and adjustments to these transactions. Administrative transactions are monetary and non-monetary transactions performed in support of financial transactions, typically by supervisory personnel. A complete list of transactions supported by the VVDH module is shown in the following tables.

Financial Transactions

The following table describes the financial transactions supported by the VVDH module. Each transaction contained in the table is followed by a short description.

FINANCIAL TRANSACTIONS	
Transaction	Description
Normal Purchase	Purchase of goods or services at the point of sale using a credit or a debit card.
Mail/Phone Order	Purchase of goods or services through the mail or a telephone call. This transaction can be initiated using a credit or a debit card.
Merchandise Return	Return of merchandise to a retailer for refund. The refund is electronically credited to the customer account. This transaction can be initiated using a credit or a debit card.
Cash Advance	Cash disbursement from a customer's account at the point of sale. This transaction can be initiated using a credit or a debit card.

FINANCIAL TRANSACTIONS	
Transaction	Description
Check Guarantee	Electronically reserves funds in a customer's checking account to cover a check written by the customer. BASE24-pos supports this transaction in pass through mode. This means BASE24-pos provides a link between the transaction and destination by simply passing the transaction to the destination without performing authorization.
Purchase Adjustment	Corrects an error in a previously-completed purchase transaction.
Merchandise Return Adjustment	Corrects an error in a previously-completed merchandise return transaction.
Cash Advance Adjustment	Corrects an error in a previously-completed cash advance transaction.
Stored Value Balance Inquiry	Balance inquiry on a stored value card.
Gift Card Balance Inquiry	Balance inquiry on a gift card.
Gift Card Purchase	Normal purchase (redemption) using a gift card.
Gift Card Return	Merchandise return using a gift card.
Gift Card Load	Replenishment of a gift card.
Gift Card Unload	Full redemption of a gift card.
Prepaid Card Activation	Activation of a prepaid card.
Prepaid Card Activation Reversal	Reversal of a prepaid card activation.
Prepaid Card Load	Replenishment of a prepaid card.
Prepaid Card Load Reversal	Reversal of a prepaid card load.

Administrative Transactions

The following table describes the batch close administrative transaction supported by the VVDH module.

ADMINISTRATIVE TRANSACTIONS	
Transaction	Description
Close Batch Request	Ends the current batch and opens a new batch to which data is logged and accumulated. When a close batch transaction is received, the system creates a batch totals record from the terminal totals in the PTDF and writes it to the PTLF. It then clears the batch totals in the PTDF.

Integration with BASE24-pos

BASE24-pos is a complete point-of-sale authorization and switching system that uses the technology of the Tandem computer to provide reliable, fault-tolerant service 24 hours a day, seven days a week.

BASE24-pos offers routing of transactions based on card type and retailer floor and ceiling limits. If the primary destination for authorization is not available, BASE24-pos allows the transaction to be sent to up to two additional authorization centers before a decision is made about approving, denying, or referring the transaction. Destinations to which transactions can be routed include host computers and interchange networks.

In addition, BASE24-pos offers a comprehensive set of authorization options on the Tandem computer. These include local and national negative identification, positive identification, and positive identification with account balances. Another authorization option for credit card accounts, parametric authorization, enables institutions to define additional criteria on which to base the authorization of transaction requests that exceed standard cardholder limits.

BASE24-pos also includes online control of all network communications elements and POS terminals, as well as a comprehensive set of daily and periodic reports for service providers and individual retailers.

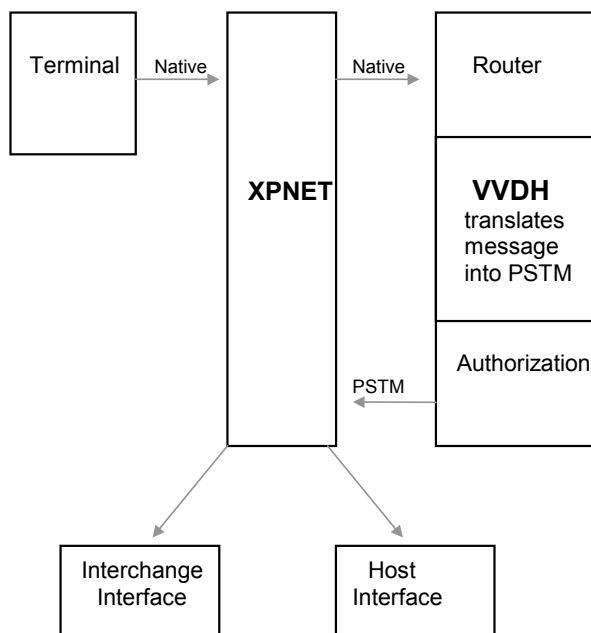
Transaction Flow

When a POS terminal sends a message to BASE24, it is first received by the XPNET process, the transaction manager, which then sends the message to the bound Device Handler/Router/Authorization process.

The VVDH module is bound into a Router/Authorization process. Any messages coming into the bound process from a POS device are first received by the Router module which then routes the message to the Device Handler module. Messages coming into the bound process from any other destination are also received first by the Router module, but are routed instead to the Authorization module to be logged to the POS Transaction Log File (PTLF).

Tying the VVDH Module to BASE24-pos

The VVDH module translates messages from the (native) VISA Second Generation record format used by the terminal into the BASE24-pos Standard Internal Message (PSTM) which is understood by all the other BASE24-pos processes. When BASE24-pos is sending a message to the terminal, the VVDH module translates the message from the PSTM format to the VISA Second Generation record format used by the terminal. The following diagram illustrates the VVDH module link. Note that the VVDH module is bound with a BASE24-pos Router/Authorization module.



Section 2

Terminal Configuration

Options supported by the VVDH module are based on the way the Base, VVDH module, and associated BASE24-pos files are set up. These files consist of the following:

- POS Retailer Definition File (PRDF)
- POS Terminal Data File (PTDF)
- POS Transaction Log File (PTLF)
- Logical Network Configuration File (LCONF)

VVDH Configuration Files

This section describes the files used to configure the VVDH module. The descriptions explain the role that each file plays in determining the options supported by the VVDH module.

The LCONF is a BASE24 file that is vital in determining VVDH module configuration issues. BASE24-pos files PRDF and PTDF also contain information set up by the customer in order to allow the VVDH module to support the necessary options. Each of these files is explained below and on the following pages.

POS Retailer Definition File (PRDF)

The POS Retailer Definition File (PRDF) contains one record for each retailer in the network. These records are used in validating POS transactions and defining authorization and routing parameters. For more information on the PRDF, refer to the *BASE24-pos Files Maintenance Manual*.

POS Terminal Data File (PTDF)

The POS Terminal Data File (PTDF) contains one record for each POS terminal in the network, including terminals that are supported by the VVDH module. The TERMINAL TYPE field located on screen 1 of the PTDF identifies the type of terminal defined in a specific PTDF record, and must be set to a value of "71" for all terminals driven by the VVDH module. The VVDH module uses the Terminal ID Number field (Format 5) or Store Number / Terminal Number fields (other formats) contained in the VISA Second Generation request message to access the appropriate record in the PTDF.

For more information on the PTDF, refer to the *BASE24-pos Files Maintenance Manual*.

POS Transaction Log File (PTLF)

The POS Transaction Log File (PTLF) contains information about the transactions processed by BASE24-pos. All approved or denied transactions are logged to the PTLF. Values in the TERMINAL CUTOVER field and the five LOG TOTALS fields on PTDF screen 3 determine if and when totals are logged to the PTLF. For more information about the PTDF and PTLF, refer to the *BASE24-pos Files Maintenance Manual*.

Logical Network Configuration File

One Logical Network Configuration File (LCONF) exists for each logical network. It contains the assigns and params associated with the network.

Assigns associate the physical names of objects in the logical network to names used in BASE24 code. Assign information must be present, otherwise BASE24 is unable to process. Most assigns link the actual physical file names to logical file names as they are referenced in BASE24 program code. For example, in order to access the Visa II Terminal Configuration File (VVCNFG), a process or server must be able to identify the fully-qualified name of the file as it exists in a logical network. The LCONF must contain an assign linking the VVCNFG name to the actual physical file. Other assigns link the physical names for different hardware to the names as they are referenced in the BASE24 program code. For example, the types of hardware referenced in BASE24 program code includes spooler locations, terminal locations, and security devices.

Params set processing values at the logical network level. In some instances when param information is not present, a process abends during initialization; however, some params are not critical for processing.

Assigns

The VVDH module requires several assigns in the LCONF in order to perform processing. These assigns are as follows and are in addition to any assigns required by the Router/Authorization module bound to the VVDH module.

POS-PTDF — Contains the name of the POS Terminal Data File (PTDF).

POS-V2DH-VVCNFG — Contains the name of the Visa II Terminal Configuration File (VVCNFG).

POS-V2DH-VVTERM — Contains the name of the Visa II Terminal Identification File (VVTERM).

Params

Several params are used by the VVDH module during processing. These params are as follows and are in addition to any params required by the Router/Authorization module bound to the VVDH module.

POS-DH-AUTH-RESP-TIMEOUT — Indicates the number of seconds a VVDH module waits for a response from the Authorization module. When a VVDH module

sends a request to the Authorization module, it also sets a timer for the number of seconds set in this param. If the timer expires before the VVDH module receives a response, the VVDH module declines the request.

If this param is not present in the LCONF or is set to an invalid value, the VVDH module logs an error message and uses a default value.

POS-DH-MAX-IN-CORE — Indicates the maximum amount of extended memory the VVDH module must reserve for keeping copies of PTDF records in memory, and, thus, determines the maximum number of transactions a VVDH module can process at one time.

If this param is set too low, the VVDH module can run out of extended memory during peak processing times, and transactions are declined until more extended memory becomes available. If the param is set too high, the excess extended memory is wasted. If this param is not present in the LCONF or is set to an invalid value, the VVDH module logs an error message and uses a default of 10.

Since the POS-DH-MAX-IN-CORE param represents the maximum number of transactions the VVDH module can process at one time, the value of the POS-DH-MAX-IN-CORE param should not be greater than the value of the POS-DH-MAX-TDFS param.

POS-DH-MAX-TDFS — Indicates the maximum number of PTDF keys that a single VVDH module can store in memory and, thus, the maximum number of terminals each VVDH module can service.

When the VVDH module is initialized or reinitialized, space is allocated in memory to build a table. This table stores the record positions of all PTDF records that the VVDH module can access. The POS-DH-MAX-TDFS param determines the table size and indicates the maximum number of terminals each VVDH module can service.

If this param is set too low, the VVDH module runs out of space in the table and is unable to process all of the POS devices. If this param is set too high, the excess table space is wasted. If this param is not included in the LCONF or is set to an invalid value, the VVDH module logs an error message and uses a default value of 20.

Section 3

VISA II Device Handler (VVDH) Module

In BASE24-pos, the Device Handler module is bound together with a Router/Authorization module to create the Device Handler/Router/Authorization process. The Device Handler/Router/Authorization process is composed of three modules—Device Handler, Router, and Authorization. Each module has distinct functions, but they operate together and are defined in the Network Environment File (NEF) as one bound process. Because the three modules are bound, transactions passed between them do not travel through the XPNET process.

The Router and Authorization modules are the same for each Device Handler/Router/Authorization process. However, the Device Handler module differs between Device Handler/Router/Authorization processes depending on the type of POS device attached to the system. For example, a different Device Handler module is required for VISA II devices than is required for Hypercom devices.

This section is devoted to describing the processing associated with the Device Handler module required for VISA II devices; the VVDH module. For more information on the Router and Authorization modules, refer to the ***BASE24-pos Router/Authorization Manual***.

This section describes the processing performed by the VVDH module. It includes the following information about the VVDH module:

- An overview of the VVDH module
- A description of each file used by the VVDH module during processing
- A description of the steps the VVDH module performs during initialization and warmbooting
- Network text commands recognized by the VVDH module
- Log message generation

VVDH Module Overview

The VVDH module is the primary interface between a POS device and BASE24-pos. Each type of POS device has its own message format and method of processing transactions. The VVDH module supports VISA II devices. The VVDH module reformats messages sent from VISA II devices into the POS Standard Internal Message (PSTM). The PSTM allows the other processes associated with BASE24-pos to process the transaction.

Likewise, all messages that BASE24-pos sends to VISA II devices must be reformatted into *native mode*, a message format the VISA II device can understand. The VVDH module performs this translation before the transaction message is forwarded to the VISA II device.

The VVDH module also controls several other functions in the BASE24-pos system. These functions are listed and briefly described in the following paragraphs.

Enforcing Transaction Sequence

The VVDH module can verify the sequence number in a message request matches the sequence number stored in the PTDF for a specific transaction. Sequence number checking is intended to ensure that BASE24-pos has received and processed each transaction only once.

Updating Accumulator Fields and Status Flags

The VVDH module updates the accumulator fields and status flags in the POS Terminal Data File (PTDF) record for each VISA II device. For example, when a transaction passes through the Device Handler/Router/Authorization process, the VVDH module updates the accumulation fields on screen 3 and 4 of the PTDF depending on the type of transaction and the processing required.

Cutting Over Terminals

The VVDH module cuts over terminals for daily settlement via a cutover transaction. The institution designates a timeframe in which cutover must take place and designates a transaction to start cutover. When the Device Handler/Router/Authorization process receives that transaction during the designated timeframe, the terminal is cut over. If the VVDH module does not perform this cutover, the Settlement Initiator performs a force cutover.

Decrypting PINs

The VVDH module can decrypt PINs via software for incoming transactions. Refer to the *BASE24 Transaction Security Manual* for more information on PIN decryption.

Controlling Response Timers

The VVDH module sets and stops transaction response timers as transactions pass in and out of the VVDH module. When the VVDH module receives a transaction from a device, it sets a response timer and then passes the transaction on for authorization. When processing completes, the VVDH module deletes the timer. If processing does not complete before the timer expires, the transaction is either declined or reversed, depending on what type of transaction timed out.

File Usage

The VVDH module uses the following files during processing. Each file listed is followed by a brief description explaining how it is used by the VVDH module.

Logical Network Configuration File (LCONF) — The assigns and params used during processing are read from the LCONF at initialization and reinitialization, and are maintained in memory.

Log Message Configuration File (LMCF) — The LMCF contains log message information such as text, routing code, and severity, that has been modified by the user. At initialization, the Device Handler/Router/Authorization process builds a table in memory from the LMCF records that contains modified information. When a log message is generated, the Device Handler/Router/Authorization process accesses the table to verify whether that log message has been modified. If the log message has been modified, the Device Handler/Router/Authorization process logs the modified information instead of the standard log message.

POS Retailer Definition File (PRDF) — The VVDH module accesses the PRDF when processing terminal cutover.

POS Terminal Data File (PTDF) — The VVDH module uses records in the PTDF to build a table of PTDF key positions in extended memory. This table enables the VVDH module to access and update the PTDF records as transactions are processed.

POS Transaction Log File (PTLF) — The PTLF contains information about the transactions processed by BASE24-pos. Approved or denied transactions are logged to the PTLF only if they are sent by a terminal. When a close batch is performed, totals are logged to the PTLF if logging PTLF totals is configured in the LOG TOTALS field on screen 3 of the PTDF. If logging PTLF totals is not configured, no totals are logged to the PTLF.

VVDH Module Initialization and Warmboot

This section describes the steps taken by the VVDH module when it is initialized or warmbooted. Understanding these steps requires knowledge of the background information provided in the following paragraph.

In BASE24-pos, Device Handler modules are bound together with Router/Authorization modules. Each of these modules has distinct functions, but they operate together and are defined in the Network Environment File (NEF) as one bound process. Because the three modules are bound, transactions passed between them do not travel through the XPNET process. In addition, the Router and Authorization modules are the same for each Device Handler/Router/Authorization process. However, the Device Handler module differs between Device Handler/Router/Authorization processes depending on the type of POS device attached to the system. For example, the HPDH module is used with Hypercom devices, and the VISA II Device Handler module is used with VISA II devices.

Initializing the VVDH Module

When the Device Handler/Router/Authorization process is initialized, the Router module begins initialization first. Part of the initialization process required by the Device Handler module is within the Router module. The steps described in this section assume the Router module has already initialized. For a complete description of the steps taken by the Router and Authorization modules when initializing, refer to the ***BASE24-pos Router/Authorization Manual***.

The VVDH module performs as follows to initialize:

1. Initializes the logging process with the name of the specific VVDH module being initialized and reads the Log Message Configuration File (LMCF).
2. Sets up a pointer used to retrieve VVDH device-dependent data from the POS Terminal Data File (PTDF).
3. Allocates an extended memory segment for the Confirmation and Batch Station to Terminal tables. If the VVDH module is unable to allocate space, the VVDH module abends.

4. Retrieves the following assigns and params from the Logical Network Configuration File (LCONF).

Assigns

POS-PTDF
POS-V2DH-VVCNFG
POS-V2DH-VVTERM

Params

POS-DH-AUTH-RESP-TIMEOUT
POS-DH-MAX-IN-CORE
POS-DH-MAX-TDFS

5. Opens the PTDF and sets the file mode so that if the VVDH module attempts to lock or read a PTDF record while any other process has the record locked, it will receive an error 73. This error indicates the file or record is locked by another process. If the VVDH module is unable to open the PTDF, the VVDH module abends.
6. Opens the Visa II Terminal Configuration File (VVCNFG) and the Visa II Terminal Identification File (VVTERM). If the VVDH module is unable to open the VVCNFG or the VVTERM, the VVDH module abends. The VVDH module sets the file mode so that if the VVDH module attempts to lock or read a record while any process has the record locked will receive an error 73. This error indicates the file or record is locked by another process.
7. Allocates extended memory space for the number of PTDF records it needs at one time. The number of records that can be in memory at one time is specified in the POS-DH-MAX-TDFS param.

Warmbooting the VVDH Module

When the VVDH module receives a WARMBOOT (9501) command, it performs the same steps described in the above initialization procedures with one exception. Step 1 in the initialization procedures described above indicates the VVDH module initializes the logging process with the name of the specific VVDH module process being initialized. This step is not performed by the VVDH module when it receives a WARMBOOT command.

Instead, when a WARMBOOT command is received, the first step taken by the VVDH module is to read the Log Message Configuration File (LMCF). An internal table stored in the VVDH module containing LMCF information requires updating if log messages have been altered via the LMCF. The VVDH module requires warmbooting in order for any LMCF changes to take effect. The logging facility itself need not be reinitialized because it was already started during initialization.

The remaining steps (beginning with step 2) described in the initialization procedures above are identical when a WARMBOOT command is received by the VVDH module.

Network Text Commands

Network text commands are used to send information from the network control facility to a process. These commands can be entered by an operator via the network control facility and serve as a tool that operators can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process, and resolving problems encountered by a specific process. The following text commands can be used with the VVDH module:

COMMKEY — Takes the key that was injected into the terminal and enters it into the VVDH module. The VVDH module can then store this key in the PTDF. This command is used when the terminal does not support BASE24 parameter downloading. If the terminal does not receive the PIN key from the VVDH module (that is, if the PIN key is not downloaded from the VVDH module), this command makes the VVDH module aware of the new key.

RESETSTATUS — Resets the transaction status in the TRAN-STATUS field of the PTDF to NO-TRAN-IN-PROGRESS. This command is intended for use if a transaction abends or some other uncontrollable event leaves the transaction status corrupted in the PTDF.

9501**** — Directs the VVDH module process to warmboot.

For a list of the steps the VVDH module follows when a WARMBOOT command is issued, refer to “Warmbooting the VVDH Module” topic. For detailed descriptions and the syntax of all text commands, refer to the *BASE24 Text Command Reference Manual*.

Log Message Generation

The VVDH process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

Log messages are contained in files on a designated Tandem subvolume (BAxxLOGM). Each file contains log messages for a specific BASE24 module. Files on the designated Tandem volume and subvolume are organized according to product module ID (PMID). One PMID exists for each process in BASE24. PMIDs identify a specific process in the BASE24 system (for example, PSV2DH identifies the VVDH module). Access to the log message documentation in file format provides customers with greater flexibility by allowing them to print out log message documentation as needed at their location or to electronically manipulate the data to suit their needs.

In addition, ACI has developed online log message documentation. The online log message documentation provides the message number, severity, and text for each log message generated by the VVDH module. In addition, it also provides the cause of each message and the recommended action to take.

For complete online log message documentation, refer to the ***BASE24 Log Message Documentation***.

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Section 4

VISA II External Message

The VISA II External Message is a variable-content, variable-length, proprietary message format developed by Visa International. The message is configured differently, based on the type of message being sent. This section contains the following information concerning the VISA II External Message:

- Standard message headers
- Optional data
- Request message requirements
- Response message
- Transaction codes

Standard Message Headers

Common to all supported VISA Second Generation request message formats are ‘headers’ comprised of three data elements.

- Record Format element
- Application Type element
- Message Delimiter element (value “. ”)

Record Format Element

Value	Description
5	Initial Second Generation Terminal format
F	Second Generation Terminal format superseding Format 5 (non-REPS CVV)
G	Direct Debit authorization format (PS/2000 REPS)
J	Credit Card and Check Guarantee format (PS/2000 REPS)
K	Credit Card and Check Guarantee format (PS/2000 CPS)
A	Direct Debit Card authorization request format
C	Debit Card Confirmation request
D	Credit Card (with Addendum Data) authorization request
T	Debit Cards or Electronic Benefits Transfer (EBT)
V	Debit Cards or Electronic Benefits Transfer (EBT) Confirmation

Application Type Element

Value	Description
0	Second generation authorization request, single request per connection (“Single-Trans”)
1	Second generation data capture request, single batch per connection
3	Second generation data capture request, multiple batches per connection
9	Second generation down line load request

Optional Data

This “optional data” discussion pertains to record formats “J”, “G”, “K”, “A”, “D”, and “T”.

Authorization Records

There are three separate data groupings for both authorization request and response messages.

Group I (Base) data fields must be present in every authorization request and response message.

Group II (Direct Debit) data fields can be included with Group I fields for Debit Card authorization request and response messages.

Group III (Addendum) data fields can be used to augment Group I fields with additional miscellaneous information as needed for authorization. Multiple group definitions, or versions, exist for Group III. The following Group III Versions are supported by the VVDH module:

- Version 001 - Commercial Card Request Indicator
- Version 007 - Card Verification Value 2 (CVV2)
- Version 008 - Fleet Fueling Card
- Version 009 - SET Electronic Commerce
- Version 029 – Extended AVS Data
- Version 034 – Card-Level Results
- Version 200 – Gift Card Information

Groups II and III are mutually exclusive and cannot be supplied without Group I data.

The following table shows which data groups are used with the various record formats.

REQUEST RECORD	RESPONSE RECORD	GROUP DESIGNATION
J	n/a	Credit Card/Check Guarantee (Group I Only)
G	n/a	Debit Card (Groups I + II)
K	L	Credit Card/Check Guarantee (Group I Only)
A	B	Debit Card (Groups I + II)
C	n/a	Debit Confirmation Record
D	E	Credit Card/Addendum Data (Groups I + III)
T	U	Debit Card or EBT (Groups I + II + III)
V	n/a	Debit Card or EBT Confirmation

Data Capture Records

The “K” format data capture records may contain one or more optional data groups appended to the required base data group. The VVDH module will process selected optional data fields from the following optional data groups if they are present in a Batch Detail record.

K-Format Batch Detail Record Optional Data Groups processed by the VVDH module	
Optional Data Group	Description
1	Cashback
2	Restaurant
5	Hotel
8	Purchasing Card

Request message requirements

The first two characters of a request message indicate the message format being used by the VISA II device and the type of processing requested.

The VVDH module supports request messages with the following combinations of record format and application type elements:

Record Format	Application Type	Description
5	0	Format 5 Authorization Request
5	1	Format 5 Data Capture Request, single batch per connection
5	9	Format 5 Down Line Load Request
F	0	Format F Authorization Request
G	0	Format G Authorization Request
J	0	Format J Authorization Request
K	0	Format K Authorization Request
A	0	Format A Authorization Request
C	0	Debit Confirmation Record
D	0	Format D Authorization Request
K	1	Format K Data Capture Request, single batch per connection
K	3	Format K Data Capture Request, multiple batches per connection
T	0	Format T Authorization Request
V	0	Format V Confirmation

Response message requirements

The structure of the response message which the VVDH module returns to the POS terminal varies depending on the type of request and the record format.

For the newer record formats which are based on the “K” format (K, A, and D), the first three bytes of a response record consist of the same data elements present in the request message (record format, application type, and message delimiter).

Response records in the older record formats (5, F, J, and G) do not begin with the record format, application type, and message delimiter. They begin with items like payment service indicator, store number, and terminal number. Refer to the *Visa External Interface Specification* documents for information about the structure and content of request and response records in these older formats.

In the response to a request in one of the record formats which are based on the “K” format, the record format is “incremented” by one character, and the application type remains the same. The following table summarizes the possible combinations.

REQUEST RECORD		RESPONSE RECORD		Description and Group Designation
Rec Frmt	Appl Type	Rec Frmt	Appl Type	
K	0	L	0	Authorization Request/Response - Group I only
A	0	B	0	Authorization Request/Response - Groups I + II
D	0	E	0	Authorization Request/Response - Groups I + III
K	1	K	1	Batch Trailer Response Record
K	3	K	3	Batch Trailer Response Record
T	0	U	0	Authorization Request/Response - Groups I + II + III
V	0	n/a	n/a	Authorization Confirmation Record

When formatting an E-format response record, the VVDH module will populate and include the optional data groups (Group III addendum data) only if applicable data exists. In other words, a request record which includes Group III Version 001 data *may not* produce a response containing Group III Version 001. (For example, the presence of Grp.III Ver.001 in a request instructs BASE24 to look for a Commercial Card Indicator. If a commercial indicator is not associated with this card, the response record will not contain Grp.III Ver.001, but *may* include Grp.III Ver.000 indicating there is no addendum data included with the response message.)

Transaction codes

The VVDH module supports authorization requests with the following transaction codes:

Authorization Transaction Codes

Tran Code	Transaction Type	Comments
01	Purchase	Standard Authorization Request
02	Mail / Phone Order	Used primarily in Direct Marketing/Mail Telephone Order Environments
03	Cash Advance	For Use in Banks and Financial Institutions Only
54	Purchase	Standard Authorization Request
55	Cash Advance	For Use in Banks and Financial Institutions Only
56	Mail / Phone Order	Used primarily in Direct Marketing/Mail Telephone Order Environments
5L	Credit/Stored Value Balance Inquiry	Balance Inquiry
71	Check Guarantee	Telecredit, Los Angeles
72	Check Guarantee	TeleCheck
73	Check Guarantee	JBS
74	Check Guarantee	Telecredit, Tampa
93	Direct Debit: Purchase	Standard Debit Authorization Request
94	Direct Debit: Purchase Return or Interlink Merchandise Credit	Debit Refund Transaction (Full or Partial Amt)
95	Interlink Debit Cancel	Must re-enter the Customer Data Field and Cardholder Identification Data (PIN)
G3	Gift Card Balance Inquiry	Remaining balance on a gift card
G4	Gift Card Purchase/Redemption	Purchase/redemption transaction on a gift card
G5	Gift Card Return/Refund	Return/refund transaction on a gift card
G6	Gift Card Add Value/Load Card	Add or load value to a gift card balance
G7	Gift Card Decrease Value/Unload Card	Decrease or unload value from a gift card balance
P1	Prepaid Card Activation	Credit type
P2	Prepaid Card Activation Reversal	Credit type
P3	Prepaid Card Load	Credit type
P4	Prepaid Card Load Reversal	Credit type

R1	Prepaid Card Activation	Debit type
R2	Prepaid Card Activation Reversal	Debit type
R3	Prepaid Card Load	Debit type
R4	Prepaid Card Load Reversal	Debit type

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Section 5

VVDH Configuration Considerations

This section contains information concerning additional decisions the customer must make when configuring the VISA II Device Handler (VVDH) module. These configuration decisions include the following:

- PIN encryption
- Sequence number checking
- Draft capture
- Transaction accumulation totals

PIN Encryption

The VVDH module supports the ANSI 9.8 PIN block. Customers are not required to support PIN encryption. The VVDH module supports PIN encryption with the DES algorithm performed in software. **The VVDH module does not support PIN encryption with the DES algorithm performed in hardware.** PIN encryption performed in software uses the ANSI PIN block, 12 right-most digits of the PAN excluding the check digit encryption method. The PIN encryption method is set in the PIN ENCRYPTION TYPE field on screen 7 of the POS Terminal Data File (PTDF). Whether a terminal can support PIN entry is indicated in the VALIDATE PIN field on screen 7 of the PTDF. For more information on the PIN fields contained in the PTDF, refer to the ***BASE24-pos Files Maintenance Manual***.

If the terminals are using record format “5”, the PIN encryption keys can be sent to the terminals in a download. In this case, PIN encryption keys are sent encrypted under the terminal master key set up in the PIN MASTER KEY field on screen 7 of the PTDF. The master key is used to encode and decode PIN keys that are transmitted between terminals and the host. The master key injected into the terminal must be the same as the master key in the PTDF.

If record format “5” is not used, the communications key is manually entered or injected into the POS device and then must be stored in the PTDF via a (9500) COMMKEY network text command.

In general, if a PIN is included in a request, it is verified by the Router/Authorization module. The VVDH module does not enforce the presence of a PIN in any request. To require that a PIN be included in a particular request, enforcement must be at the terminal level or Router/Authorization module level using the Card Prefix File (CPF) configuration. For more information concerning the CPF, refer to the CPF documentation contained in the ***BASE24 Base Files Maintenance Manual***. For an in-depth discussion of PINs, refer to the ***BASE24 Transaction Security Manual***.

Sequence Number Checking

Sequence number checking is intended to ensure that BASE24-pos has received and processed each transaction only once. When the VVDH module receives a request, the VVDH module verifies that the sequence number is the one the VVDH module is expecting. The VVDH module checks the sequence number sent in the request with the sequence number stored in the PTDF.

The sequence number stored in the BASE24-pos Standard Internal Message (PSTM) is made up of a prefix generated by the VVDH module followed by the four byte transaction sequence number from the POS device.

Draft Capture

BASE24-pos allows for processing retailer transactions as draft capture or non-draft capture. Non-draft capture transactions are transactions for which the paper receipt represents the actual settlement instrument. Paper receipts must be physically presented for deposit in order for the transaction to be settled. While the transaction could have been authorized by BASE24-pos, the BASE24-pos record cannot be used to settle the transaction.

Draft capture transactions are electronically captured at the time of the transaction and are sufficient in themselves to enact a funds transfer for the transaction amount. BASE24-pos identifies draft capture transactions in its files and includes draft capture totals on its retailer reports so that the retailer sponsor can settle with the retailer.

Draft capture is a settlement tool only. It has no impact on the authorization of transactions. Cardholders are still checked against the same authorization criteria without regard for whether a transaction is draft capture or non-draft capture. Thus, the retailer is not incurring additional risk of cardholder fraud by using or not using draft capture.

Authorization with Electronic Follow-up

The method of draft capture used in the VVDH module is authorization with electronic follow-up. The VVDH object must be built with the Data Capture Upload Module (VVDC) in order to use data capture.

The customer should set the TP field on PTDF screens 8, 9, and 10 to indicate the authorization with electronic follow-up (TP = 2) draft capture option for each terminal and card type.

The authorization with electronic follow-up option provides for a two-step transaction that first obtains online authorization for the amount of a customer transaction and then, in a second step, transmits the information needed to perform settlement to the card issuer. The two steps are described below.

1. Obtain an online authorization for the amount of a customer transaction. If the transaction is authorized, the second step in this option is performed. If the customer transaction request is declined, the second step is not taken.

BASE24-pos logs these initial transactions to the POS Transaction Log File (PTLF) with a draft capture flag of 2, indicating the transaction has been authorized but requires an electronic follow-up transaction.

2. If the transaction amount has been authorized in step 1, collect the ticket information electronically and transmit it to the card issuer for settlement.

BASE24-pos logs the second transaction to the PTLF with a draft capture flag of 3, indicating that this is the electronic follow-up to a previously approved transaction. Electronic follow-up transactions update the draft capture totals in the PTDF.

Setting the Draft Capture Flag in the PSTM

The draft capture flag indicates whether BASE24-pos balances should immediately reflect transaction activity. The Router module sets this draft capture flag using the TRAN-PROFILE field in the POS Standard Internal Message (PSTM).

The VVDH module uses the option specified in the TP (transaction profile) field on PTDF screens 8, 9, and 10 to set the TRAN-PROFILE field in the PSTM. The Router module then moves the value in the TRAN-PROFILE field of the PSTM to the DFT-CAPTURE-FLG field in the PSTM. Valid values for the DFT-CAPTURE-FLG field are as follows:

- 0 = Authorize only and use a paper receipt to settle the transaction later
- 1 = Authorize and draft capture; transaction includes the information needed for immediate settlement
- 2 = Authorize only and expect electronic follow-up to settle the transaction later
- 3 = Electronic follow-up to settle a previously authorized transaction

The following table shows the transaction profile values and the corresponding draft capture flag settings used by the VISA II Device Handler module.

TP	DFT-CAPTURE-FLG	Transaction Type
2	0	Card verification transactions
	2, 3	All transactions except card verification transactions

Note: VISA II draft capture with electronic follow-up is a two-transaction process and 0220 messages are sent in batch mode.

For more information about draft capture processing refer to the ***BASE24-pos Transaction Processing Manual*** and the ***BASE24-pos Settlement and Reporting Manual***.

Transaction Accumulation Totals

The BASE24-pos VVDH module supports balancing terminals by batch. The terminal's batch-close settlement request contains a detail record for each transaction in the batch as well as a trailer record containing batch accumulator counts and totals. The VVDH module returns the status of the batch to the terminal in a batch response record. If the terminal totals disagree with the VVDH module totals in the PTDF, they are logged to the PTLF.

The VVDH module always accumulates batch totals. Upon a batch close for the corresponding period, the VVDH module records batch totals in the PTLF and resets the batch totals. All totals are cleared if a close transaction is entered any time within the retailer's cutover window set in the POS Retailer Definition File (PRDF).

The BALANCE SUPPORT field on PTDF screen 4 indicates to the VVDH module which totals are supported by the terminal by means of close transactions. This field must be set to 0 or 9 for VISA II devices. A 0 in this field indicates that only batch totals are supported and a 9 indicates that no totals are supported by the terminal. Totals specified in the PTDF are allowed to accumulate until reset when a batch close transaction is entered. Other totals are reset at the logical network cutover.

Refer to the *BASE24-pos Files Maintenance Manual* for more information about the PTDF totals screens. Refer to the *BASE24-pos Settlement and Reporting Manual* for more information about how PTDF totals are accumulated.

Section 6

Transaction Message Flows

This section contains examples of the message flows specific to the VVDH module. Each example includes a diagram illustrating the message flow, followed by a description of each step. The VVDH module transaction message flows listed below are documented in this section. For more information on general BASE24-pos transaction message flows, refer to the BASE24-pos Transaction Processing Manual.

- Approved normal purchase received by the device
- Approved normal purchase; communications between device and BASE24 down
- Declined normal purchase received by the device
- Declined normal purchase; communications between the device and BASE24 down
- Batch Close Settlement Transaction - Good Batch
- Batch Close Settlement Transaction - Rejected Batch

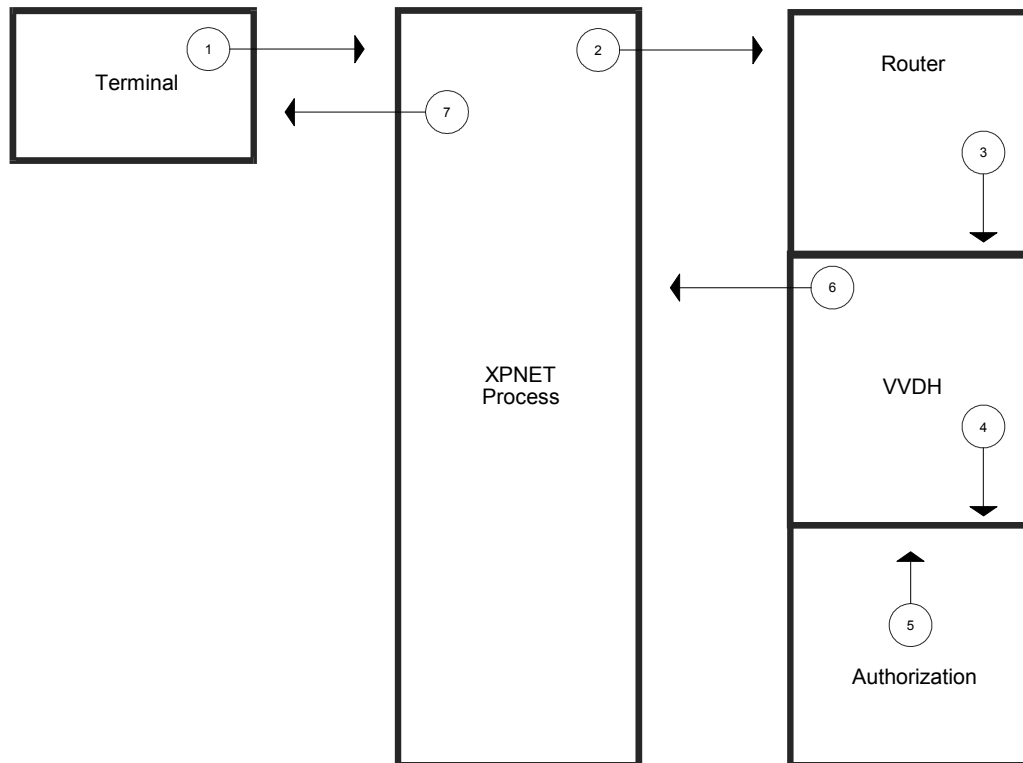
When a terminal sends a message to BASE24, it is received by the XPNET process, which forwards the message to the BASE24 VISA II Device Handler (VVDH) module.

The VVDH module is bound into a BASE24-pos Router/Authorization process. Any messages coming into the bound process from a BASE24 Standard POS Device are first received by the Router module which then routes the message to the Device Handler module.

The VVDH module is fully compatible with release 6.0 of BASE24-pos. The following diagrams illustrate the exchange of messages between the VVDH module and the terminal during transaction processing using BASE24-pos. The data shown is assumed to follow the standard message header and to be terminated with an ETX in each situation.

Approved Normal Purchase Received by Device

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction with a response. In this example the transaction was approved.



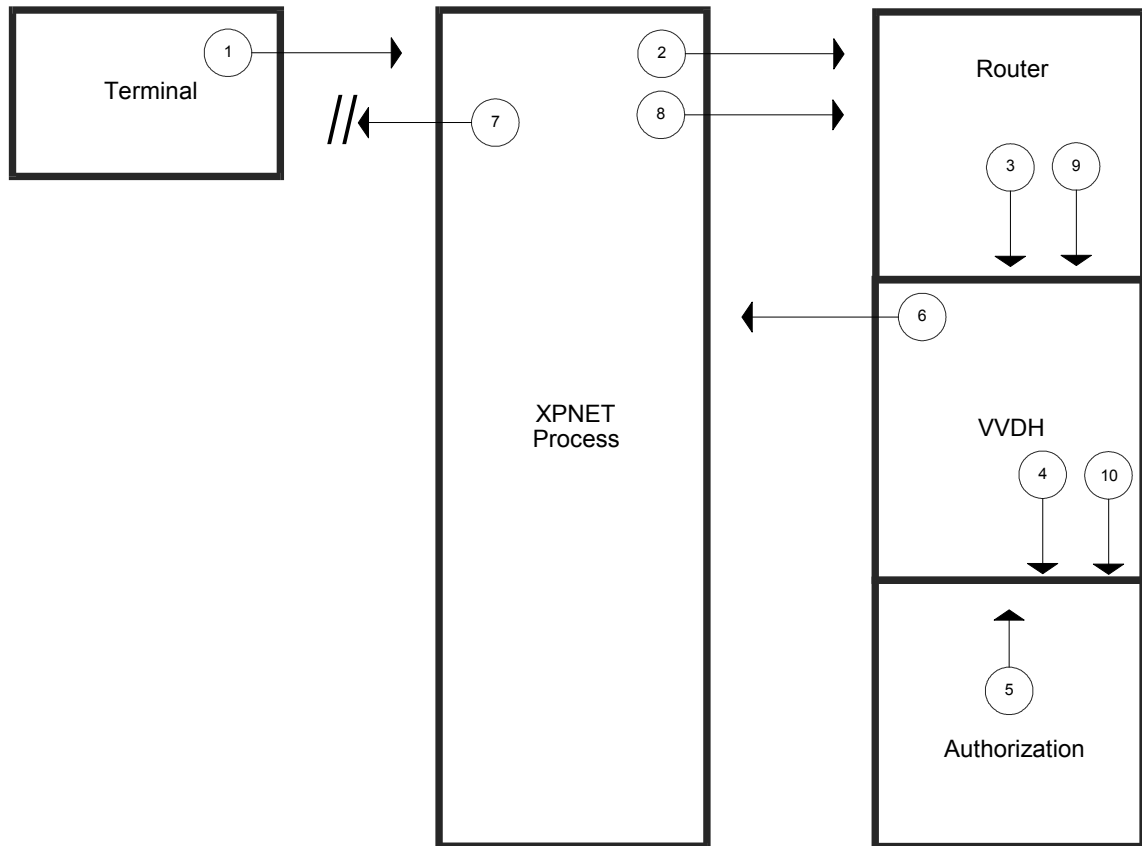
Steps

Processing

- 1, 2, 3 The terminal sends its native message for a normal purchase to the VVDH module. This message passes through the XPNET process and the Router module.
- 4 The VVDH module validates the request and formats the message in a POS Standard Internal Message (PSTM) for a normal purchase. The VVDH module then sends the message to the Authorization module.
- 5 The Authorization module returns a response to the VVDH module indicating the transaction was approved.
- 6, 7 The VVDH module updates the PTDF totals, translates the response from PSTM format into the terminal's native language, and sends the response to the terminal. The native mode response travels though the XPNET process.

Approved Normal Purchase; Communications between Device and BASE24 Down

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction that was not received by the device because communications between the device and BASE24 was down. In this example, the transaction was



reversed.

Steps

Processing

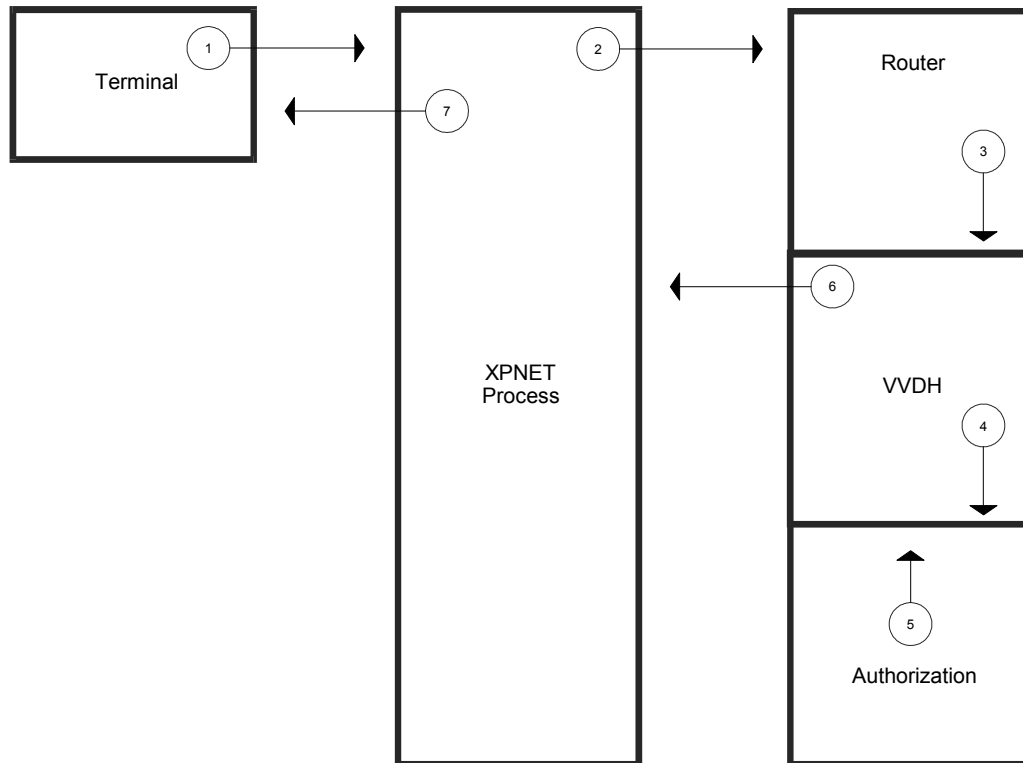
- 1, 2, 3 The terminal sends its native message for a normal purchase to the VVDH module. This message travels through the XPNET process and the Router module.
- 4 The VVDH module validates the request and formats the message in a POS Standard Internal Message (PSTM) for a normal purchase. The VVDH module then sends the message to the Authorization module.

Steps	Processing
5	The Authorization module returns a response to the VVDH module indicating the transaction was approved.
6, 7	The VVDH module updates the PTDF totals, translates the response from PSTM format into the terminal's native language, and sends the response to the terminal via the XPNET process.
8, 9	The XPNET process attempts to send the response to the terminal, but is unsuccessful. The XPNET process returns the failed message to the VVDH module. This message travels through the Router module.
10	The VVDH module generates a reversal transaction and sends it to the Authorization module.

When formatting the reversal after validating, the VVDH module formats the message in PSTM format again using the context for a reversal. The VVDH module sends the reversal to the Authorization module and updates PTDF totals.

Declined Normal Purchase Received by Device

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction with a response. In this example the transaction was declined.



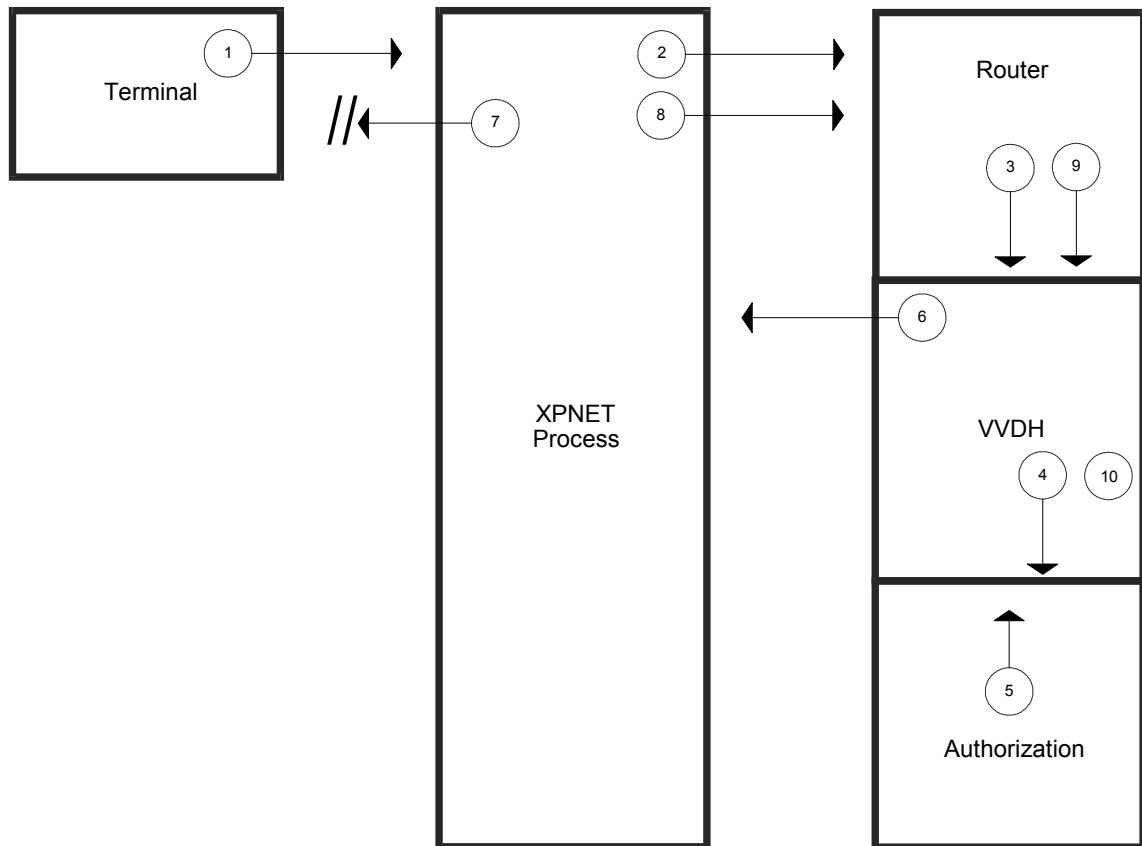
Steps

Processing

- 1, 2, 3 The terminal sends its native message for a normal purchase to the VVDH module. This message passes through the XPNET process and the Router module.
- 4 The VVDH module validates the request and formats the message in a POS Standard Internal Message (PSTM) for a normal purchase. The VVDH module then sends the message to the Authorization module.
- 5 The Authorization module returns a response to the VVDH module indicating the transaction was declined.
- 6, 7 The VVDH module translates the response from PSTM format into the terminal's native language and sends the response to the terminal. This message travels through the XPNET process.

Declined Normal Purchase; Communications between Device and BASE24 Down

The diagram below illustrates the transaction message flow in a scenario containing a normal purchase transaction that was not received by the device because communications between the device and BASE24 were down. In this example, the transaction was declined.



Steps

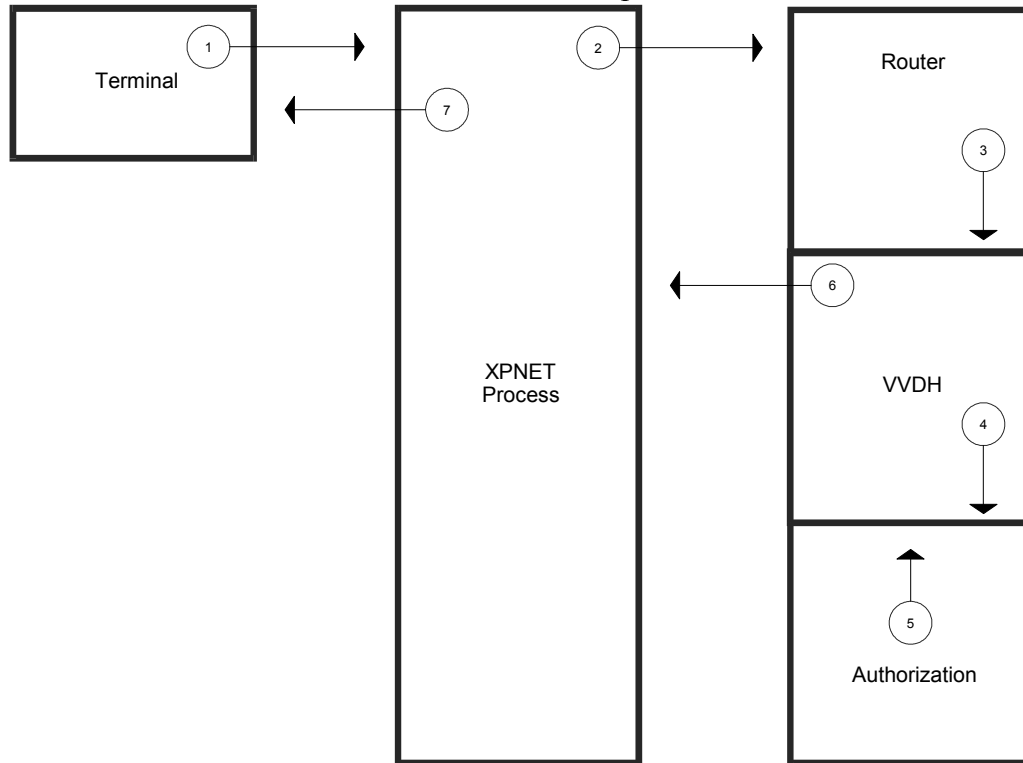
Processing

- 1, 2, 3 The terminal sends its native message for a normal purchase to the VVDH module. This message travels through the XPNET process and the Router module.
- 4 The VVDH module validates the request and formats the message in a POS Standard Internal Message (PSTM) for a normal purchase. The VVDH module then sends the message to the Authorization module.

Steps	Processing
5	The Authorization module returns a response to the VVDH module indicating the transaction was declined.
6	The VVDH module translates the response from PSTM format into the terminal's native language and sends the response to the terminal via the XPNET process.
7	The XPNET process attempts to send the response to the terminal, but is unsuccessful. The device times out accordingly.
8, 9	The XPNET process sends the failed message to the VVDH module. This message travels through the Router module.
10	The VVDH module updates the PTDF totals again.

Batch Close Settlement Transaction - Good Batch

The diagram below illustrates the transaction message flow in a scenario containing a batch close settlement transaction. In this example, the batch is closed successfully.



Steps

Processing

- 1, 2, 3 The terminal sends its native message (sequence) for a batch close transaction to the VVDH module. This message travels through the XPNET process and the Router module.

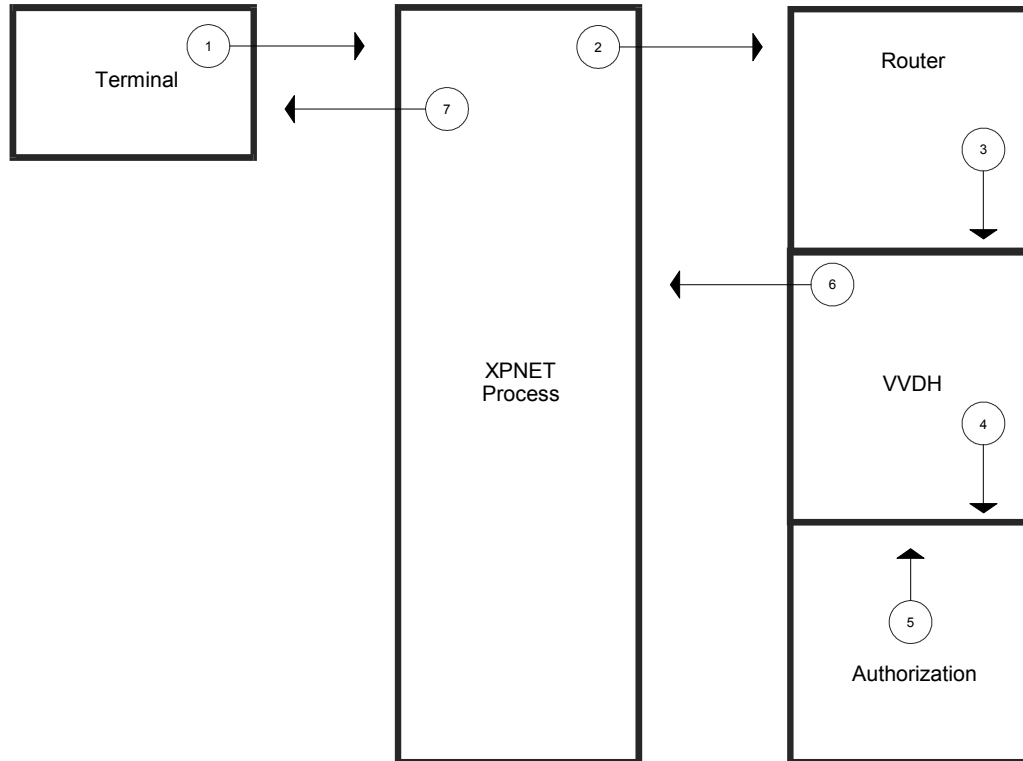
There will be a sequence of messages coming from the terminal during a batch close. (header record, parameter record, a detail record for each transaction in the batch, and a trailer record) The XPNET process acknowledges each message from the terminal, but the VVDH module does not respond to the terminal until after processing the trailer record.

- 4 For each detail record received from the terminal, the VVDH module verifies the record is not a duplicate, then formats and sends a PSTM completion message to the Authorization module.

Steps	Processing
5	The Authorization module receives the completion message and posts it to the cardholder's account. The VVDH module updates its batch accumulation totals.
6, 7	The VVDH module's batch accumulation totals match the terminal's totals included in the trailer record, so a batch response record is formatted with a response code of "GB" (good batch) and sent through the XPNET process to the terminal. The batch totals are logged to the PTLF. The VVDH module determines whether the terminal should be cut over to the next posting date based on the cutover parameters set in the POS Retailer Definition File (PRDF). The VVDH module logs PTDF totals by card type (service totals) to the POS Transaction Log File (PTLF). If the VVDH module is cutting the terminal over, network totals are logged to the PTLF, and the posting date is increased by one.

Batch Close Settlement Transaction - Rejected Batch

The diagram below illustrates the transaction message flow in a scenario containing a batch close settlement transaction. In this example, the VVDH module rejects the terminal's batch close request.



Steps

Processing

- 1, 2, 3 The terminal sends its native message (sequence) for a batch close transaction to the VVDH module. This message travels through the XPNET process and the Router module.

There will be a sequence of messages coming from the terminal during a batch close. (header record, parameter record, a detail record for each transaction in the batch, and a trailer record) The XPNET process acknowledges each message from the terminal, but the VVDH module does not respond to the terminal until after processing the trailer record, unless there is a reason to reject the batch before receiving the trailer record.

- 4 For each detail record received from the terminal, the VVDH module verifies the record is not a duplicate, then formats and sends a PSTM completion message to the Authorization module.

Steps	Processing
5	The Authorization module receives the completion message and posts it to the cardholder's account. The VVDH module updates its batch accumulation totals.
6, 7	The VVDH module's batch accumulation totals do not match the terminal's totals included in the trailer record, or the batch header, parameter, detail, and trailer records were received out of sequence, or some other error occurred while processing the batch close request. A batch response record is formatted with a response code of "RB" (rejected batch) and sent through the XPNET process to the terminal.

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Appendix A

American Express Standard Industry Formats

The AMEX Data Collection option allows customers to perform draft capture on transactions originating from American Express cardholders. Generally, American Express has defined categories under which transactions are placed. These categories are as follows:

- Auto rental
- Lodging
- Restaurant
- General retail
- Oil

For each of these categories, American Express requires different data to be sent from the device. Therefore, American Express has set forth standard industry formats for each category to ensure the data they require is captured by the device and sent to BASE24. American Express supports a Descriptive Billing File (DBF) that is used to submit charges to American Express without submitting paper. The format of the DBF must conform to the American Express standard.

The VVDH module is able to provide the gratuity information required in the restaurant standard industry format and, subsequently, capture and process transactions sent using this format. The VVDH module fills the fields for which it cannot provide information with blanks (or zeros for amount fields). It is then the responsibility of the host to determine if additional information is required to build the DBF.

In conjunction with this file, the customer must set up Standard Industrial Classification (SIC) Codes or Merchant Category Codes in the PTDF. SIC and Merchant Category Codes identify the merchant's line of business. Refer to the ***BASE24-pos Files Maintenance Manual*** for more information about the PTDF.

The VVDH module's use of the American Express Standard Industry Formats is limited. When the VVDH module receives a data capture detail record (during a batch close transaction) that contains optional data group 2 (restaurant gratuity), it will set the American Express Industry Format Code to 12 (restaurant) and populate the TIP1-AMT

field with the gratuity amount. All other fields in the Restaurant transaction category are initialized to spaces or zeros.

The VVDH module does not use the auto rental, lodging, general retail, or oil transaction categories.

The standard industry format for the restaurant transaction category is shown in the following table.

Restaurant Standard Industry Format

Field	Length	Description
frmt-cde	9(2)	The American Express Industry Format Code identifies which standard industry format should be used for a particular transaction. This field contains a value of 12 for restaurant transactions.
reference-cde	9(9)	A reference number used by American Express to identify the transaction and obtain supporting data from the service establishment for a charge. Valid values are 000000000 through 999999999.
charge-cde1	X(4)	A code mapped by the host to a description that identifies the transaction associated with the charge. This value is defined by American Express.
tab-roc-id	X(10)	The original Record of Charge (ROC), invoice, or another number used to identify the original transaction. This is an internal American Express code.
description-cde	X(1)	A code that describes the nature of the transaction. Valid values are as follows: 0 = Food 1 = Food and beverage 2 = Gift certificate 3-9 = User-defined A-Z = User-defined
tax-amt	9(18)	The amount of sales tax charged for the transaction. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.

Field	Length	Description
food-amt	9(18)	The total cost of food, or food and beverage if these charges are combined on the original record of charge (ROC). This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0.
beverage-amt	9(18)	The total cost of beverages, if itemized in the DESCRIPTION-CDE field of the American Express token. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0. Refer to the BASE24 Tokens Manual for additional information about this token.
tip1-cde	X(1)	A code mapped by the host to a description that designates an employee receiving a tip.
tip2-cde	X(1)	A code mapped by the host to a description that designates a second tip itemized on the original record of charge (ROC).
tip1-amt	9(18)	The total amount of the tip described in the TIP1-CDE field of the American Express token. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0. Refer to the BASE24 Tokens Manual for additional information about this token.
tip2-amt	9(18)	The total amount of the second tip described in the TIP2-CDE field of the American Express token. This field has an implied decimal two positions from the rightmost character. If not used, this field contains 0. Refer to the BASE24 Tokens Manual for additional information about this token.

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Index

A

Address Verification, 1-9
American Express standard industry formats, A-1
Assigns. *see* Logical Network Configuration File (LCONF)
Authorization transaction codes, 4-8

B

BASE24-pos settlement and cutover, 1-7
Batch close
 administrative transaction, 1-14
 message flow for a good batch, 6-8
 message flow for a rejected batch, 6-10

D

Descriptive Billing File (DBF), A-1
Device Handler/Router/Authorization process, 1-15
Draft capture
 defined, 5-4
 setting flag in PSTM, 5-5
Draft capture option
 authorization with electronic follow-up, 5-4
Draft Capture Option, 1-7

I

Initialization, 3-5

L

Log Message Configuration File (LMCF), 3-4
Log message documentation, 1-8
Log message generation, 1-8, 3-9
Logical Network Configuration File (LCONF), 2-3, 3-4
 assigns, 2-3
 params, 2-3

M

Multiple language support, 1-6

N

Network text commands, 3-8
Nucleus process terminology change, xiii

O

Optional data, 4-4

in an E-format response record, 4-7
in Data Capture records, 4-5

P

Params. *see* Logical Network Configuration File (LCONF)
PIN encryption, 5-2
Port usage tracking, 1-10
POS Retailer Definition File (PRDF), 2-2, 3-4
POS Terminal Data File (PTDF), 2-2, 3-4
POS Transaction Log File (PTLF), 2-2, 3-4
PS2000 support, 1-9

R

Request message requirements, 4-6

S

Sequence number checking, 1-8, 3-2, 5-3
Standard message headers, 4-2
Station ID token, 1-10

T

Terminal Balancing, 1-8
Terminal cutover, 3-2
Track 1 support, 1-9
Track 2 support, 1-9
Transaction accumulation totals, 3-2, 5-6
Transaction Flow, 1-15
Transaction message flows, 6-1
Transaction Support, 1-12

V

VISA external message formats, 1-6
VISA II Device Handler (VVDH) module
 overview, 1-2, 3-2
Visa Payment Service 2000 support. *see* PS2000 support

W

Warmboot, 3-5

X

XPNET process, xii

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